FRFD: Feature-Refined and Disentangled Representations for Robust Multimodal Ophthalmic Disease Grading

Feature Refinement & Representation Disentanglement Framework for 2D Fundus and 3D OCT

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https://github.com/senweiliang-coder/cs1470final-wishtoday/tree/main

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1 Introduction

Ophthalmologists often use multiple imaging modalities because fundus photographs and OCT volumes provide complementary evidence. Fundus images show the retinal surface, optic disc, and vascular appearance, while OCT volumes reveal depth-wise retinal structure. In practice, however, complete multimodal data is not guaranteed. A patient may lack one exam, one modality may be low quality, or different clinics may have different imaging equipment. A robust model should therefore make reliable predictions with both modalities, with only one modality, and under noisy-modality corruption.

Two representation problems motivate FRFD. First, each modality contains **task-irrelevant redundancy**: fundus images include background or non-diagnostic regions, and OCT volumes contain many slices that may be weakly related to disease grade. Second, multimodal features can become **overlapped and entangled**: common disease evidence and modality-specific signals may collapse into the same latent space. When this happens, the classifier may fail to use unique fundus/OCT information, especially when one modality is missing.

FRFD addresses these problems by combining feature refinement, disentanglement, and self-distillation. KFR reduces redundancy by pulling features toward class-guided anchors. FDE separates the representation into common and unique components, aligning the common features while decorrelating modality-specific features. Self-distillation then encourages missing/noisy-modality predictions to remain close to the complete-modality branch.

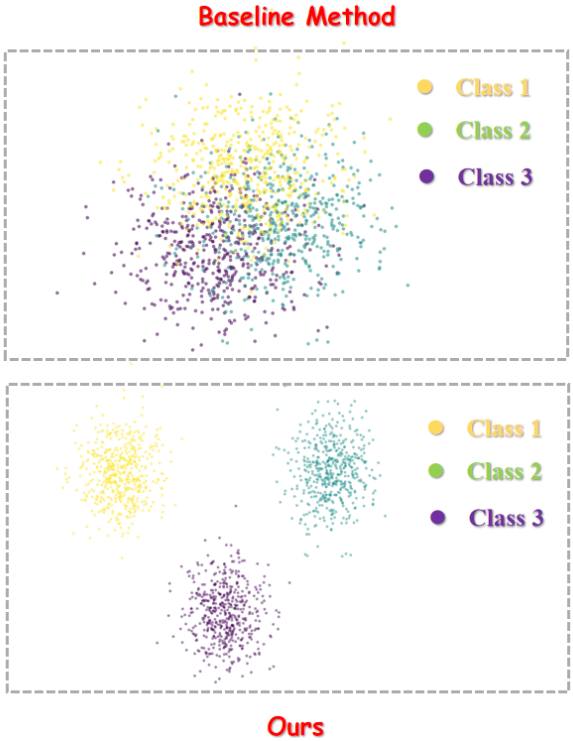

Figure 1: Motivation for FRFD. Multimodal ophthalmic inputs may contain task-irrelevant redundancy within each modality, while naive fusion can lead to overlapping and entangled fundus–OCT representations.

2 Literature Review

Multimodal ophthalmic learning. Deep learning models for ophthalmology commonly use fundus and OCT because the modalities describe different aspects of the eye. A standard baseline extracts a fundus feature with a 2D encoder and an OCT feature with a 3D encoder, then concatenates them for classification. In our reproduction, the fundus branch follows the vision-transformer idea of ViT [1], and the OCT branch follows a 3D transformer-style encoder similar to UNETR [2].

Fusion and attention. Early fusion and late fusion are simple but can preserve redundant or noisy features. Attention-based fusion, such as CBAM-style channel/spatial attention [3], can improve feature weighting; our comparison includes M^2LC as this type of multimodal attention baseline. These methods help when both inputs are available, but they do not directly enforce robustness when a modality is missing.

Missing-modality learning and disentanglement. Missing-modality methods try to preserve performance when only partial information is observed. IMDR is used as a stronger baseline because it targets incomplete multimodal ophthalmic grading [6]. FRFD follows the broader idea that robust representations should separate what is shared across modalities from what is unique to each modality. We also use self-distillation and distribution matching: the complete-modality branch guides incomplete branches at both feature and logit levels, with losses related to product-of-experts reasoning and maximum mean discrepancy [4, 5].

3 Methodology

3.1 Problem Formulation

Let the dataset be $\mathcal{D} = \{(\mathbf{x}_i^F, \mathbf{x}_i^O, y_i)\}_{i=1}^N$, where $\mathbf{x}_i^F$ is a fundus image, $\mathbf{x}_i^O$ is an OCT volume, and $y_i \in \{1, \ldots, C\}$ is the disease grade. The model must predict under three settings: complete modality, single-modality missingness, and noisy-modality corruption. Modality encoders produce

$$\mathbf{z}_i^F = E_F(\mathbf{x}_i^F), \qquad \mathbf{z}_i^O = E_O(\mathbf{x}_i^O). \tag{1}$$

Instead of directly classifying $[\mathbf{z}_i^F; \mathbf{z}_i^O]$, FRFD first refines and disentangles the features.

3.2 FRFD Overview

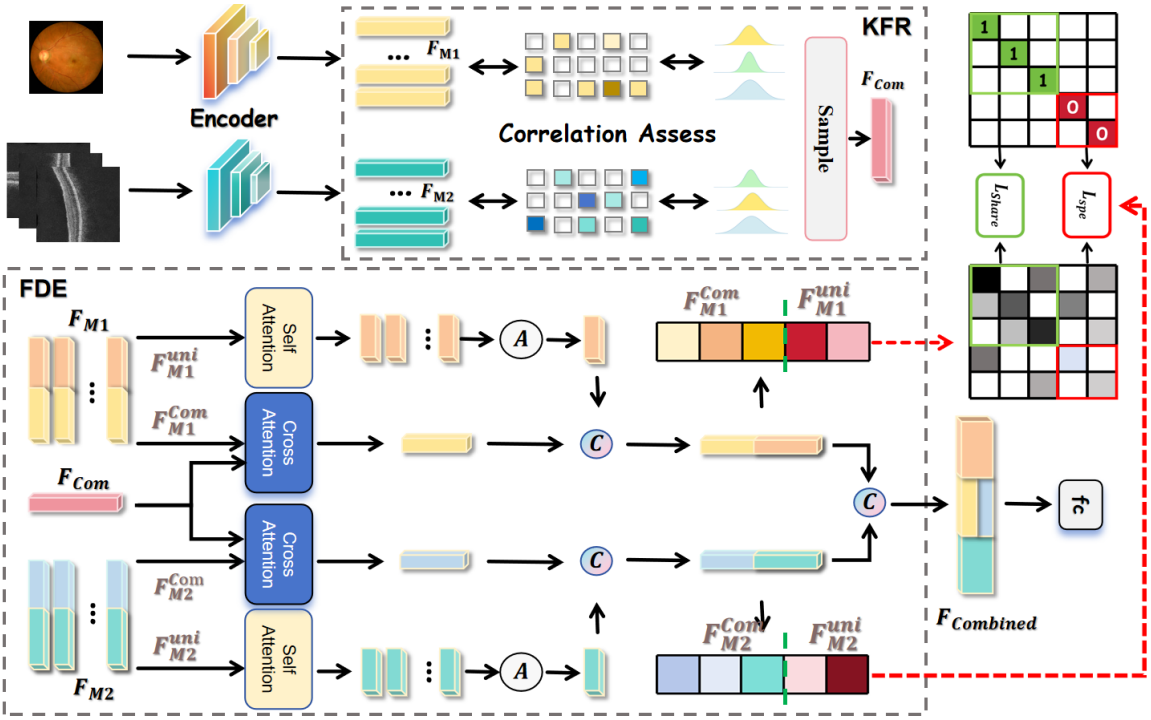

Figure 2: Overview of the proposed FRFD framework. KFR performs class-guided feature refinement, FDE decomposes features into common and modality-unique components, and self-distillation regularizes incomplete-modality predictions using the complete-modality branch.

The full framework has two main modules. **KFR** performs class-guided feature refinement with anchors and guiding tokens. **FDE** decomposes the refined representation into modality-common and modality-unique components. The

recombined feature is passed to the classifier, while auxiliary losses enforce anchor matching, common-feature alignment, unique-feature decorrelation, and self-distillation.

3.3 KFR: Class-Guided Feature Refinement

For each modality $m \in \{F, O\}$ and class c, KFR maintains a learnable feature anchor $\mathbf{a}_{m,c}$. With cosine similarity $s(\cdot,\cdot)$, the anchor-matching loss is

$$\mathcal{L}_{\text{matching}} = -\frac{1}{2N}\sum_{i=1}^{N}\sum_{m\in\{F,O\}}\left[s(\mathbf{z}_i^m, \mathbf{a}_{m,y_i}) - \frac{1}{C-1}\sum_{c\neq y_i} s(\mathbf{z}_i^m, \mathbf{a}_{m,c})\right]. \tag{2}$$

This encourages features from the correct disease grade to move toward the correct anchor and away from other class anchors. During inference, the model uses correlation assessment to select the most relevant anchor because labels are unavailable. The selected anchor provides a guiding signal so that the model focuses on disease-relevant regions or OCT slices rather than redundant visual content.

3.4 FDE: Feature Disentanglement

FDE splits each refined feature into a common part and a unique part:

$$\mathbf{h}_i^m = [\mathbf{h}_i^{m,\text{com}}; \mathbf{h}_i^{m,\text{uni}}], \qquad d_c = pd, \quad d_u = (1-p)d. \tag{3}$$

The common subspace should contain disease evidence shared by fundus and OCT; the unique subspace should retain modality-specific information. FDE measures fundus/OCT dependence with a batch-wise cross-correlation matrix

$$C_{ij} = \frac{\sum_b \mathbf{h}_{b,i}^F \mathbf{h}_{b,j}^O}{\sqrt{\sum_b (\mathbf{h}_{b,i}^F)^2}\sqrt{\sum_b (\mathbf{h}_{b,j}^O)^2}}. \tag{4}$$

For common features, C should approach the identity matrix; for unique features, it should approach zero. The losses are

$$\mathcal{L}_{\text{com}} = \sum_i (1 - C_{ii}^{\text{com}})^2 + \lambda_c \sum_i \sum_{j\neq i} (C_{ij}^{\text{com}})^2, \tag{5}$$

$$\mathcal{L}_{\text{uni}} = \sum_i (C_{ii}^{\text{uni}})^2 + \lambda_u \sum_i \sum_{j\neq i} (C_{ij}^{\text{uni}})^2. \tag{6}$$

The final feature $\mathbf{h}^{\text{combined}}$ is built by recombining aligned common features and decorrelated unique features.

3.5 Self-Distillation and Full Objective

To stabilize missing/noisy-modality predictions, FRFD uses the complete-modality branch as a teacher-like reference. Feature consistency is enforced by maximum mean discrepancy,

$$\mathcal{L}_{\text{features}} = \text{MMD}(\mathbf{h}_{\text{miss}}^{\text{combined}}, \mathbf{h}_{\text{complete}}^{\text{combined}}), \tag{7}$$

and logit consistency is enforced by Jensen–Shannon divergence,

$$\mathcal{L}_{\text{logits}} = D_{\text{JS}}(p_{\text{miss}} \,\|\, p_{\text{complete}}). \tag{8}$$

The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{matching}} + \mathcal{L}_{\text{com}} + \mathcal{L}_{\text{uni}} + \mathcal{L}_{\text{features}} + \mathcal{L}_{\text{logits}}. \tag{9}$$

4 Results

4.1 Setup and Main Results

We evaluate on three Harvard-30k ophthalmic grading tasks: age-related macular degeneration (AMD), diabetic retinopathy (DR), and glaucoma. Metrics are accuracy (ACC), AUC, and F1-score [7]. We compare with a simple transformer-fusion baseline, B-IF, M^2LC, and IMDR. "OCT" and "Fundus" indicate single-modality testing, while "Both" indicates complete-modality testing [**?**].

Table 1 shows that FRFD has the best overall performance across AMD, DR, and glaucoma. The complete-modality setting benefits from both modalities, while the single-modality columns show that the learned representation remains useful when fundus or OCT is absent.

Table 1: Performance comparison on Harvard-30k ophthalmic disease grading tasks. "3D" denotes OCT-only inference, "2D" denotes fundus-only inference, and "Both" denotes complete multimodal inference.

Method	Metric	AMD			DR			Glaucoma		
		3D	2D	Both	3D	2D	Both	3D	2D	Both
Baseline	ACC	65.07	72.92	70.87	70.53	73.81	74.07	65.69	73.02	73.35
	AUC	69.88	75.38	81.06	69.94	79.11	78.73	69.86	75.35	74.53
	F1	69.64	72.28	70.83	62.01	70.46	71.17	70.91	72.31	71.64
B-IF	ACC	69.57	72.35	73.17	69.05	73.62	76.36	69.64	73.39	73.39
	AUC	70.14	71.98	83.82	65.25	67.50	77.95	68.95	76.61	73.32
	F1	67.45	70.03	71.25	67.93	69.68	75.61	67.18	72.47	72.11
M^2LC	ACC	68.97	73.24	74.93	67.20	73.04	75.21	67.70	72.78	74.98
	AUC	72.23	72.67	82.39	65.05	67.89	79.68	71.22	70.23	76.45
	F1	65.06	73.80	71.20	64.33	74.59	74.39	65.60	71.11	74.23
IMDR	ACC	70.62	75.17	79.50	72.62	76.19	78.57	71.16	75.54	77.31
	AUC	72.69	80.48	85.09	74.69	79.07	85.00	75.07	78.47	78.98
	F1	71.90	76.59	72.52	72.90	72.18	77.04	70.37	75.12	78.90
	ACC	**71.79**	**76.69**	**81.42**	**74.38**	**77.50**	**79.50**	**72.53**	**76.28**	**78.55**
	AUC	**74.84**	**81.55**	**85.82**	**76.88**	**80.60**	**86.71**	**76.28**	**79.59**	**79.32**
	F1	**72.94**	**76.79**	**78.93**	**74.28**	**76.71**	**79.81**	**72.54**	**76.62**	**80.54**

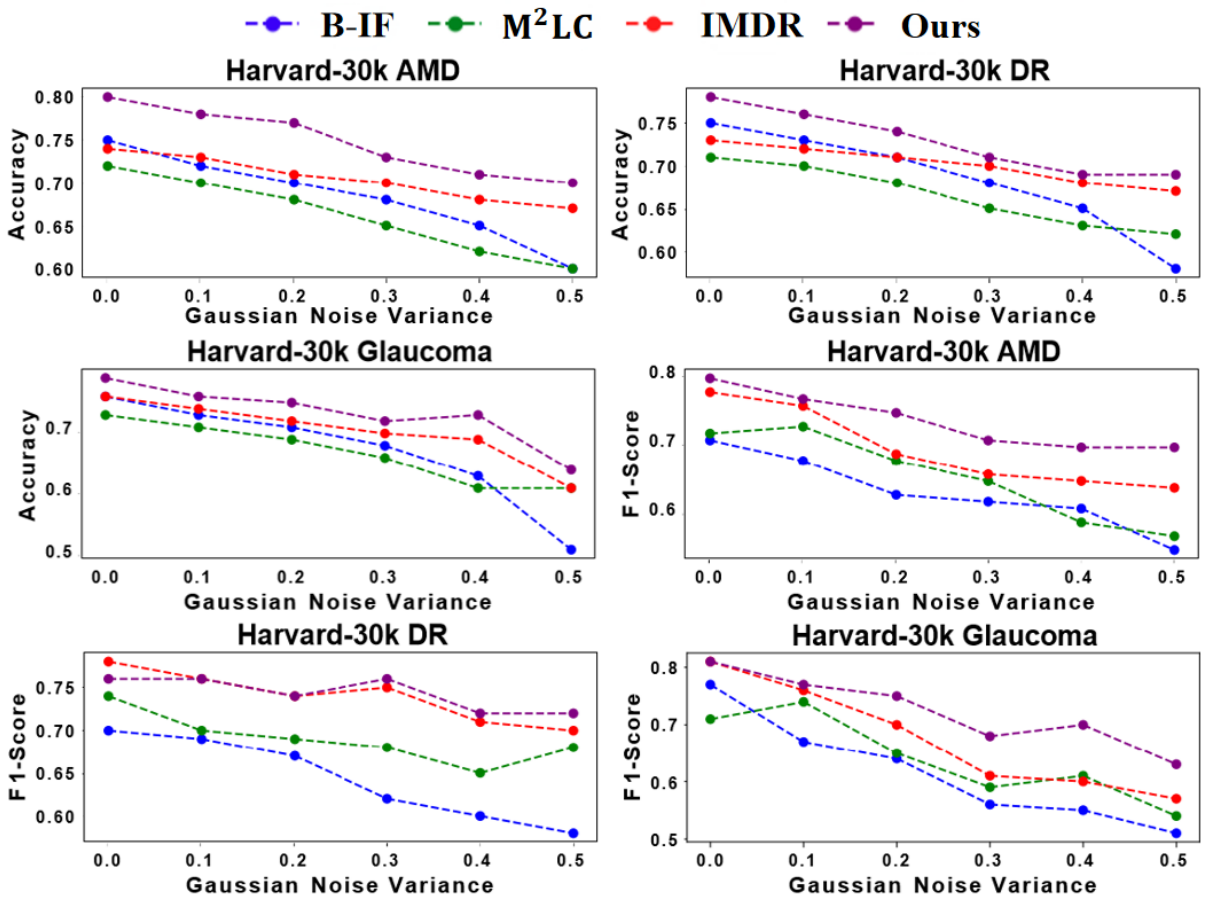

Figure 3: Robustness under noisy-modality evaluation. The curves summarize ACC and F1 trends under increasing Gaussiannoise levels across AMD, DR, and glaucoma tasks.

The noisy-modality experiment shows that all methods degrade as Gaussian noise increases, but FRFD remains stronger across AMD, DR, and glaucoma. This supports the idea that KFR filters task-relevant features and FDE keeps common and modality-specific information more separable.

4.2 Ablation, Sensitivity, and Visualization

Table 2: Ablation and shared-feature sensitivity analysis. The module ablation reports the contribution of KFR and FDE, while the sensitivity study evaluates the effect of the common-feature ratio p

(a) Module ablation.

Variant	Baseline	KFR	FDE	ACC	AUC	F1
I	✓			59.51	63.42	53.47
II	✓	✓		66.58	70.95	66.19
III	✓		✓	64.67	66.45	64.88
IV	✓	✓	✓	**69.37**	66.39	57.94

(b) Common-feature ratio p (ACC).

p	0.3	0.4	0.5	0.6	0.7
AMD	79.56	**81.42**	79.05	80.23	73.47
DR	76.87	78.13	78.75	**79.50**	77.50
Glaucoma	77.32	**78.55**	77.44	76.35	77.50

The ablation indicates that KFR improves feature selection, FDE improves feature separation, and the full model achieves the highest ACC. The sensitivity table shows that the common-feature ratio p must be balanced: too little common space weakens cross-modality sharing, while too much common space may suppress modality-specific evidence.

5 Challenges

The first challenge was the heterogeneous input structure: fundus is 2D, while OCT is a 3D volume with many slices. Direct concatenation can overemphasize redundant OCT slices or background fundus regions. KFR was introduced to make the feature selection more class-relevant.

The second challenge was evaluating missing modality fairly. A model that performs well with both modalities can fail when only fundus or only OCT is available. We therefore tested complete, OCT-only, fundus-only, and noisy-modality settings. This makes the results more realistic, but it also makes training harder because the representation must support multiple input patterns.

The third challenge was balancing multiple losses. Classification, anchor matching, common alignment, unique decorrelation, feature distillation, and logit distillation can compete. The ablation and sensitivity results show that different metrics may prefer different balances, so careful tuning is necessary.

6 Ethics

Medical AI must be handled carefully. First, patient imaging data requires privacy safeguards, de-identification, controlled access, and appropriate data-use approval. Second, robustness to missing modality does not automatically imply fairness. If a dataset overrepresents certain patient groups, scanners, hospitals, or disease severities, the model may perform worse on underrepresented groups. A clinical version should report subgroup results and uncertainty, not only aggregate ACC, AUC, and F1.

Third, FRFD should be considered a decision-support tool rather than a replacement for ophthalmologists. Wrong predictions could affect referral or treatment decisions. The model should therefore provide confidence information, quality warnings, and interpretable visual evidence where possible. Finally, our Gaussian-noise tests approximate low-quality modalities but do not fully represent real clinical missingness; future validation should include external datasets and real acquisition failures.

7 Reflection

This project showed us that robust multimodal learning is not just about combining more data. Fundus and OCT are complementary, but they also introduce redundancy, noise, and modality-specific biases. The updated FRFD naming clarifies the two main ideas: **KFR** refines features toward class-relevant anchors, and **FDE** structures the latent space into common and unique components.

We also learned that robustness claims require multiple evaluation settings. Reporting only complete-modality performance would hide whether the model works when a patient lacks one imaging modality. By testing complete, missing, and noisy settings, we obtained a better picture of model behavior. In future work, we would test real low-quality scans, calibrate prediction confidence, and evaluate fairness across demographic and clinical subgroups.

Overall, FRFD demonstrates that feature refinement, representation disentanglement, and self-distillation can improve multimodal ophthalmic grading under missing-modality conditions, while also making the model design easier to interpret.

References

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